

Forecasting Generating Power of Sun Tracking PV Plant using Long-Short Term Memory Neural Network Model: A case study in Ninh Thuan - Vietnam

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Abstract—This article provides a reliable very short-term forecast (nowcasting) of the power generation capacity of PV power plants. Specifically, in this article, a deep learning method utilized the long-short term memory model (LSTM) to predict PV system's power generation has been proposed using the Python programming language with TensorFlow library. In addition, other deep learning models such as recurrent neural network (RNN) and convolution neural network (CNN) are also used and compared with LSTM to demonstrate the advantages of LSTM compared to those models. To verify, the article used actual data collected from Ninh Thuan solar power plant - Vietnam. The results show the prediction ability with promising quality of the LSTM model.

Index Terms—PV power nowcasting, deep learning, LSTM.

I. INTRODUCTION

Electric power generation consists of power plants using nonrenewable energy sources or traditional fossil fuels with limited reserves and renewable energy sources (RES) with endless supply. Fossil fuels have been utilized over the years to generate electrical power. However, since these sources of energy give high atmospheric pollution, cause negative effects on global warming, and have a finite reserve, in the present age, their availability is reducing rapidly and many countries are focusing on RES [1]. From the literature review, the most popular RES currently are solar, wind, hydro, tidal and geothermal heat. Among all of RES, PV is a more useful

for its absence of noise generation, low maintenance costs, simplicity of design and environmental friendliness [2].

In contrast, photovoltaic (PV) energy source also has several significant negative impacts. Since solar power is an unstable resource, which depends on cloud cover, irradiation, ambient conditions, etc. so this causes uncertainty of generating energy over time. The power generated from a PV power plant can rapidly vary throughout the course of a day [1], [3]. The significance of PV generating power forecast derived from the intermittent and variable characteristics of PV power. The accuracy of predictions are crucial for ensuring the safety, stability, and economic efficiency of power grids [4].

The forecast accuracy changes with the prediction future interval if the parameters of forecast model are kept the same. Therefore, we should select the suitable forecast time horizon while designing a forecasting model. The forecast of PV power generation can be categorized into four types depends on the length of time horizon. Table I presents four-time horizons classification for PV power: very short-term

TABLE I: PV power forecasting for the time interval

PV power forecasting	Interval
Very-short term	A few seconds to 30 minutes
Short-term	30 minutes to one day
Medium-term	One day to one month
Long-term	More than one month

(from a few seconds to 30 minutes ahead), short-term (from 30 minutes to a day ahead), medium-term (from a day ahead to a month ahead), and long-term (beyond a month ahead). Very short-term forecasts are applied for voltage management and grid stability procedures. Meanwhile, short-term forecasts are employed for voltage control, economical load dispatch, load increase and decrease decisions, power reserve management, day-ahead electricity market and generator on-and-off decisions. Medium-term forecasts are utilized for unit commitment decisions and maintenance plans. Finally, long-term forecasts are considered to be conducted for wind farm design and restructured electricity market [5].

There are four primary categories of PV power forecasting techniques: physical, statistical, artificial intelligence (AI) and hybrid [6]. Physical models use numerical data of ambient conditions or satellite images for solar radiance and PV generation predicting. To employ this method, stable ambient conditions are required [6], [7]. The statistical techniques makes predictions by training with historical measured data of PV power generation. The model uses some historical data to test the trained model. Continues or ends training based on the relationship between predicted and actual data. AI based models, which can be also classified into the category of statistical approach, utilize machine learning techniques and artificial neural network to build up PV power forecasters. Finally, hybrid techniques are combination of the three aforementioned method [6], [8]. The prediction performance of physical, statistical, AI and hybrid forecasting models were reviewed and evaluated in [8]. Based on this thorough review, hybrid models demonstrate superiority when compared to models solely employing either machine learning or mathematical techniques.

Due to the distinctive architecture of LSTM featuring memory blocks that absent in other neural architectures, delivers more accurate forecasts compared to other neural network forecasting models [9], [10]. The model proposed in [11] employs both the attention mechanism and LSTM for single-step ahead forecasting of PV generation, taking into account sampling resolutions ranging from 7.5 minutes to 60 minutes. For forecasting models rely mainly on the time series of PV generation, single-step ahead forecasting can obtain a reasonable accuracy [10]. However, in terms of *multi-step ahead* forecasting, relying only on PV generation time series may produce poor forecasting efficiency. Therefore, besides the time series data of PV generation, time series features of radiation and environmental temperature are also used to improve the accuracy of the LSTM *multi-step forecasting* model.

In this study, a LSTM deep learning model is developed for *very short-term* solar PV power forecasting, also known as *nowcasting*. The LSTM architecture exhibits a heightened capacity to generate highly accurate very short-term PV power forecasts across multiple seasons of the year. Notably, it showcases remarkable efficacy in producing precise predictions even amidst turbulent weather conditions that engender chaotic radiance patterns. Additionally, for the purpose of comparative

analysis, other forecasting models such as *convolutional neural network* (CNN) and *recurrent neural network* (RNN) are included in investigation alongside the proposed LSTM model for very short-term PV generation forecasts.

The paper is organized as follows. Section II describes the processing of the data set considered to develop and train the LSTM forecasting models along with the significance of the selected data. Section III illustrates the forecasting framework based on LSTM models. Obtained results and comparisons are discussed in Section IV, followed by conclusion in Section V.

II. ANALYZE AND EVALUATE DATA

The data of PV power plant located in Ninh Thuan province, Vietnam was selected for study. These data were processed with Python Pandas and NumPy libraries.

A. Approach overview

The initial stage of data preparation involves gathering PV generation power and weather data for the specified PV plant. The data-set consist of solar radiance (W/m^2), temperature ($^{\circ}\text{C}$) and PV generation power (W) at DC side of inverter which are minutely collected. The data used for training and testing the LSTM forecasting model was gather from Ninh Thuan province, Vietnam for the entire 2020 year.

Before data is processed for PV power forecasting, missing data and outliers must be cleared from the data set for accuracy. Missing data can affect analysis results, so these missing values need to be identified and handled. There are many methods for handling missing data such as filling in the mean value, median value or using prediction models to fill in missing values. Outliers are completely different data values, which can be negative values, NaN values or values that are far different from reality. Identifying and handling outliers can be done using methods such as checking standard deviation, data distribution, and determining thresholds to remove outliers to ensure the accuracy of the data. Python is a popular and powerful tool for data processing and visualization, with libraries such as NumPy, Pandas and Matplotlib supporting this work.

B. Sliding window and Single-steps ahead forecasting

Time series data can be framed as a form of supervised learning. By restructuring a sequence of numbers from a time series dataset, it is possible to present it as a supervised learning problem. This restructuring involves utilizing previous time steps as input variables and designating the subsequent time step as the output variable. The count of previous time steps employed is commonly referred to as the window width or input width. The *sliding window* technique forms the foundation for transforming any time series dataset into a supervised learning problem. In sliding window methodology, a consistent training size is maintained while the window moves along the dataset to generate numerous train-test pairs. The way a sliding window work is clearly shown in Fig. 1. This may be a good strategy if the model is fixed size. It's also

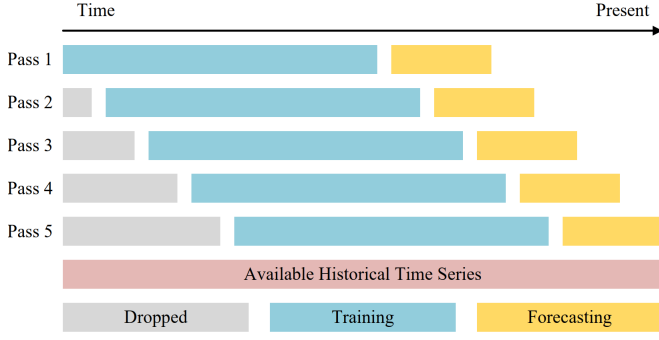


Fig. 1: Sliding Window method

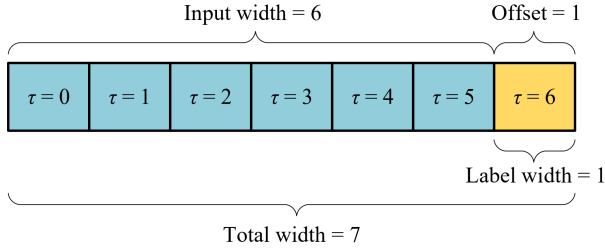


Fig. 2: Single step ahead prediction, horizon width = 1

good for initial model development as it will result in faster model builds.

Moreover, the significance lies in determining the number of time steps ahead to forecast. Similarly, it is customary to assign distinct names to the problem based on the number of time steps to be forecasted:

- *Single-step ahead* forecast: This is where the next time step ($\tau + 1$) is predicted.
- *Multi-step ahead* forecast: This is where two or more future time steps are to be predicted.

Fig. 2, Fig. 3 and Fig. 4 show the windows for single-step and multi-step forecasting.

For now, we are focusing on framing single-step ahead forecast using the sliding window method.

III. LSTM BASED FORECASTING FRAMEWORK

A. LSTM structure

An RNN architecture called LSTM is made to recognize long-term dependencies in sequential data [11]. RNNs are neural networks that analyse time series or natural language input sequentially. LSTM networks are composed of *memory cells* that may selectively store and output information from previous time steps, allowing them to manage long-term dependencies efficiently. In Fig. 5., the structure of LSTM is visualized [12]. Which data is removed from the *memory cell* is decided by the *forget gate*. In (1), the *sigmoid* function (σ) receives data from the current input $x(\tau)$ and the hidden state $h(\tau - 1)$. The sigmoid function yields a value between 0 and 1, where a value of 0 indicates that the data is entirely lost and a value of 1 indicates that the data is retained. Which data is allowed to enter the memory cell is decided by the input gate. The second sigmoid function in (2) is given two arguments:

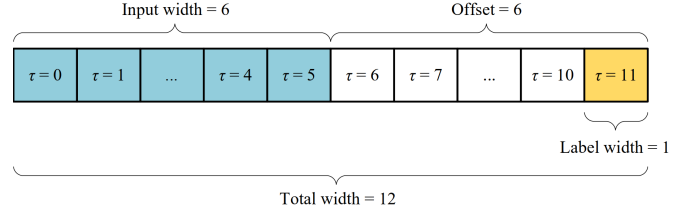


Fig. 3: Single-step ahead prediction, horizon width = 6

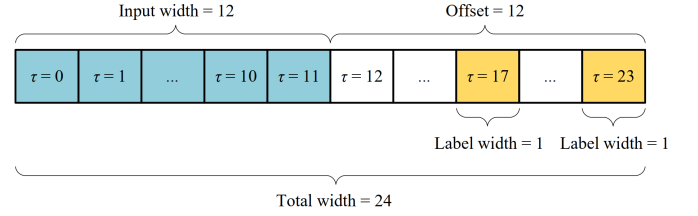


Fig. 4: Multi-step ahead prediction, horizon width = 12

the hidden state $h(\tau - 1)$ and the present state $x(\tau)$. From 0 for important to 1 for not important, the values are altered.

The *tanh* function (φ) takes input from the current state as well as the hidden state in (3) and (4). To control the network, the *tanh* operator creates a vector $c(\tau)$ that includes every potential value across -1 and 1. After that, the activation functions' output values are ready for element-wise multiplication.

By combining data from earlier inputs, the output gate in (5) and (6) determine the next hidden state. The current state and the previous hidden state are inputs to the third sigmoid function, which starts the process. After that, the new cell state that was created from the original cell state is subjected to the *tanh* function's operation. These two processes' results are compounded together. The network uses this value to determine what information the hidden state should retain. This hidden state is used to make predictions, and then the new cell state and hidden state are sent to the following time step.

$$f(\tau) = \sigma [W_f x(\tau) + U_f h(\tau - 1) + b_f] \quad (1)$$

$$i(\tau) = \sigma [W_i x(\tau) + U_i h(\tau - 1) + b_i] \quad (2)$$

$$\tilde{c}(\tau) = \phi [W_c x(\tau) + U_c h(\tau - 1) + b_c] \quad (3)$$

$$c(\tau) = f(\tau) \odot c(\tau - 1) + i(\tau) \odot \tilde{c}(\tau) \quad (4)$$

$$o(\tau) = \sigma [W_o x(\tau) + U_o h(\tau - 1) + b_o] \quad (5)$$

$$h(\tau) = o(\tau) \odot \phi(c(\tau)) \quad (6)$$

The introduction of the sign " $\odot$ " indicates the element-wise multiplication.

B. The framework for forecasting PV power

Fig. 6 illustrates the LSTM forecasting model. It utilizes historical PV power generation and weather data, including solar radiance and temperature, along with time of day and month of the season indices, as input features. The initial dataset is linearly scaled through min-max normalization, maintaining the connection between the original data's values.

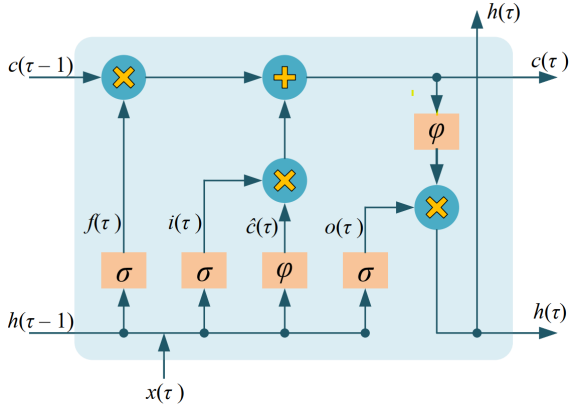


Fig. 5: Diagram depicting the internal structure of an LSTM memory block

The width of the input time sequence is chosen to be M , and the forecast horizon is selected to be H . The arrangement of the input array at time step $(\tau - 1)$ is as follows:

- The historical data of past M minutes contains PV power generation (P), solar radiance (R), and temperature (T). The sequences of them are concatenated to form an array $W \in \mathbb{R}^{M \times 3}$ as (7)

$$W = \{w(\tau - M), w(\tau - M + 1), \dots, w(\tau - 1)\} \quad (7)$$

- The sequence of time indices within the past M minutes of the day is $\{i(\tau - M), i(\tau - M + 1), \dots, i(\tau - 1)\}$ which will be transformed into an array $I \in \mathbb{R}^{M \times 24}$ by one-hot encoding because every hour of the day is encoded across 24 categories.
- The sequence of month indices of the season for past M minutes is $\{s(\tau - M), s(\tau - M + 1), \dots, s(\tau - 1)\}$ which will be encoded to a $M \times 3$ array $S \in \mathbb{R}^{M \times 3}$ since each month of a season is categorized into only three distinct categories.
- Finally, all sequences of historical data and one-hot encoded ones are concatenated to create the input array $X(\tau - 1) = \{x(\tau - M), x(\tau - M + 1), \dots, x(\tau - 1)\} \in \mathbb{R}^{M \times 30}$.

The double-layers LSTM is fed with the input array. The initial hidden layer comprises N_1 cells, whereas the second hidden layer contains N_2 cells. From each update, the cell output vector, $h \in \mathbb{R}^{N_2}$, is transmitted to the output layer for the computation of the output value $Y(\tau) = p(\tau + H - 1)$. Ultimately, $Y(\tau)$ undergoes denormalization to derive the predicted PV power generation sequence.

C. Error Metrics

The developed model's feedback is represented via error metrics. These metrics are used to create the model, and their performance is assessed and adjusted as needed. The proposed model utilizes two error metrics to assess forecasting

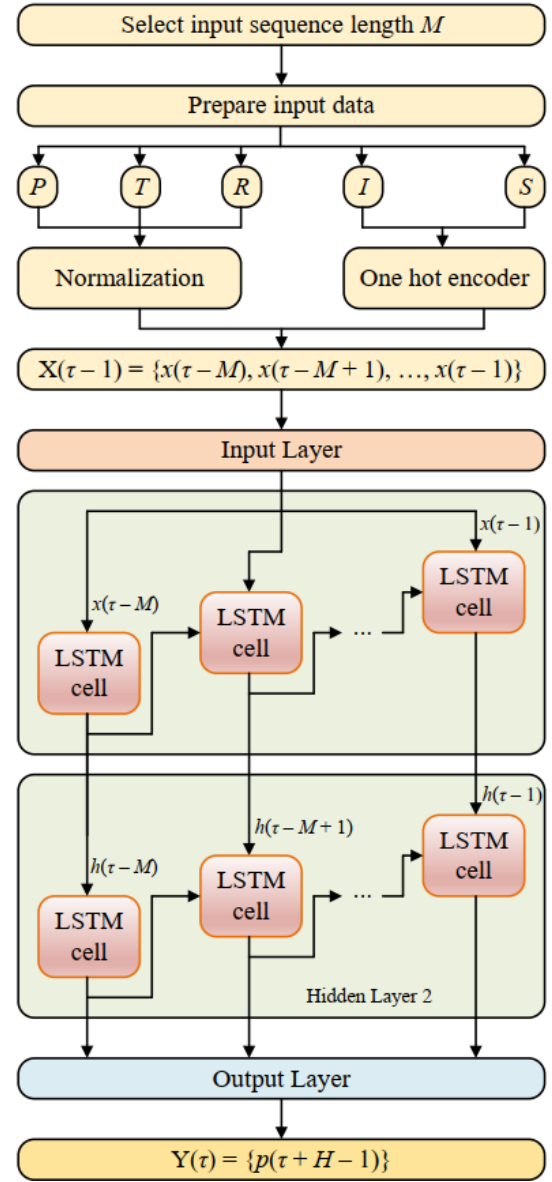


Fig. 6: LSTM forecasting framework

performance: the mean absolute error (MAE) and the root mean square error (RMSE), expressed as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_{\text{predicted}}[i] - y_{\text{actual}}[i]| \quad (8)$$

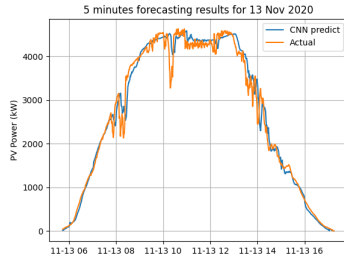
$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{\text{predicted}}[i] - y_{\text{actual}}[i])^2} \quad (9)$$

where $y_{\text{predicted}}$ is the forecasted value of the model and y_{actual} is the actual value.

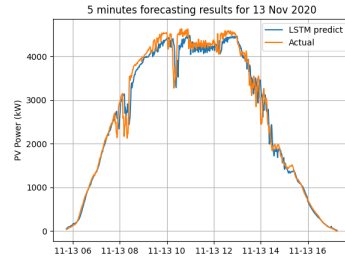
IV. RESULTS AND DISCUSSION

A. Predicting performance of proposed model

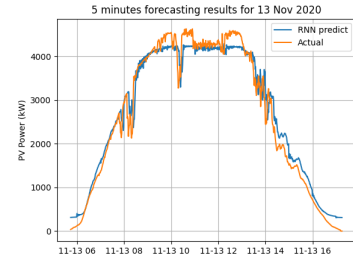
The data utilized in the experiments was collected in 2020. It is partitioned into two subsets with a training-to-testing



(a) CNN model

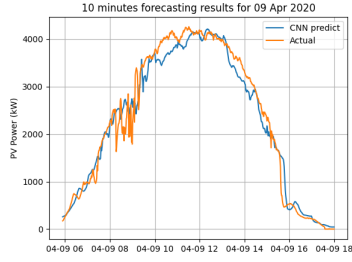


(b) LSTM model

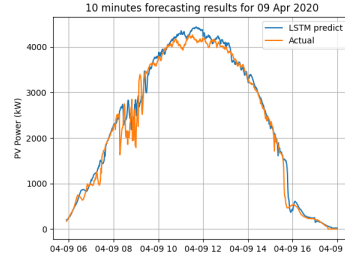


(c) RNN model

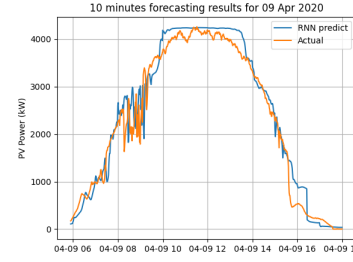
Fig. 7: 5-minute single step ahead prediction performance of CNN, LSTM and RNN models



(a) CNN model

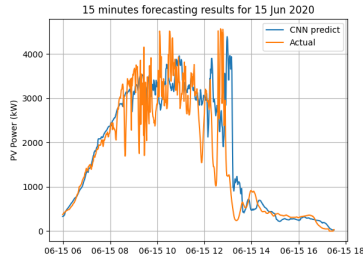


(b) LSTM model

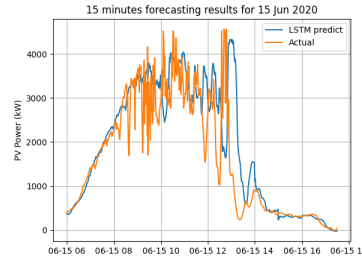


(c) RNN model

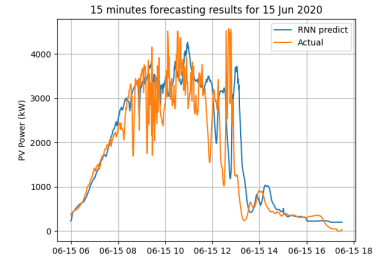
Fig. 8: 10-minute single step ahead prediction performance of CNN, LSTM and RNN models



(a) CNN model



(b) LSTM model



(c) RNN model

Fig. 9: 15-minute single step ahead prediction performance of CNN, LSTM and RNN models

TABLE II: Prediction accuracy of LSTM model for different seasons

Season	Synthetic					
	MAE (MW)	RMSE (MW)	MAE (MW)	RMSE (MW)	MAE (MW)	RMSE (MW)
	5m	5m	10m	10m	15m	15m
Spring	0.188	0.403	0.235	0.471	0.304	0.559
Summer	0.155	0.327	0.209	0.422	0.268	0.532
Autumn	0.216	0.383	0.289	0.507	0.404	0.654
Winter	0.210	0.446	0.265	0.501	0.301	0.553

ratio of 9:1. In the LSTM model, the first hidden layer comprises 100 memory cells, and the second layer comprises

TABLE III: Prediction accuracy of LSTM model for different horizon length over test set

Minute ahead	Synthetic	
	MAE (MW)	RMSE (MW)
5	0.192	0.393
10	0.249	0.477
15	0.320	0.577

50 memory cells. The algorithm is implemented in Python 3.10 with Tensorflow libraries. The proposed model's forecasting performance is assessed across various prediction horizon lengths during different seasons. The forecasting accuracy of

TABLE IV: Prediction accuracy comparison among different forecasting models in test set

Model	Synthetic					
	MAE	RMSE	MAE	RMSE	MAE	RMSE
	(MW) 5m	(MW) 5m	(MW) 10m	(MW) 10m	(MW) 15m	MW 15m
LSTM	0.192	0.393	0.249	0.477	0.320	0.577
RNN	0.265	0.423	0.287	0.481	0.303	0.527
CNN	0.206	0.389	0.260	0.478	0.318	0.540

the proposed model is then compared in Table II and Table III.

As can be seen from Table II, the prediction accuracy of the proposed forecasting algorithm varies seasonally. In spring and summer, forecast accuracy is quite good with low error. However, during autumn and winter, forecast accuracy drops significantly, due to strong fluctuations in generation capacity during these seasons. The use of LSTM model has improved the ability to forecast PV power generation, however the accuracy is still lower in autumn and winter due to strong fluctuations in power generation. The prediction accuracy of the proposed LSTM model also varies with the prediction horizons. It can be seen clearly in Table III, the model achieves the highest accuracy for 5-minute horizon while dropping to lowest with 15-minute interval. Fig. 7b, Fig. 8b and Fig. 9b show the predicting performance among different forecasting horizons in specific days in 2020.

B. Evaluation of predicting performance of forecasting models

Several NN models have been developed for the purpose of forecasting PV power. The prediction capability of the proposed LSTM model is evaluated by implementing the proposed framework using RNN and CNN. The minutely data of the test set is used to compare the performance of the forecasting models. The prediction accuracy is compared in Table IV. The predicting performance of the three algorithms considered for comparison are shown in Fig. 7, Fig. 8, and Fig. 9. From these comparison results, it can be seen that LSTM model shows the superiority over RNN and CNN when it has the highest accuracy predicting performance.

V. CONCLUSION

This paper has proposed and built a short-term PV power forecasting model using the LSTM NN together with Python programming language and the TensorFlow library to achieve this goal. Prediction accuracy is improved through testing and selecting the activation function, optimization method, number of hidden layers, number of neurons in each layer, and training/test set ratio appropriate to the data set. During the training process, the mean square error (RMSE) evaluation standard is used to compare and select the most optimal model. The forecast results in the article show positive results. Using LSTM networks in the model helps capture complex rela-

tionships in time series data, ensuring flexibility and accurate forecasting.

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